

MD ADNAN

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SUMMARY

First-year AI & Cyber Security student with a hands-on, competition-tested track record — four hackathon/CTF wins, a rigorously tested fraud-detection system with real adversarial security testing, and a disciplined 300-day public ML learning streak. Comfortable across the ML pipeline (classical ML through deep learning in PyTorch), applied system security (concurrency, prompt-injection defense, audit-trail design), and the product layer (FastAPI/React). Recently selected as an Official Contributor for Open Source Connect India 2026.

TECHNICAL SKILLS

Languages: Python, JavaScript, Java

AI/ML: PyTorch (neural networks, from-scratch), scikit-learn (Isolation Forest, Gradient Boosted Trees ensembles), YOLOv8, MediaPipe, OpenAI Whisper, TF-IDF & semantic embeddings (sentence-transformers), unsupervised learning (K-Means, DBSCAN, hierarchical clustering), statistical analysis (confidence intervals, Bayesian base-rate projection), model evaluation (cross-validation, GridSearchCV), pandas/NumPy

Backend/Web: FastAPI, React, Node.js, SQLite, Supabase (Postgres + Auth), Docker, MCP (Model Context Protocol) tool integration

Applied Security: CTF competition experience (cryptography, digital forensics, OSINT, web exploitation); concurrency/race-condition debugging, prompt-injection defense, audit-trail design, Pydantic schema validation

Tools: Git/GitHub, Vercel, Netlify

PROJECTS

Sentinel — Coordinated Abuse-Ring & Fraud-Spike Detector — *Razorpay AI Buildathon 2026, Track 02: AI Risk Manager* · [Repo](#)

- Built an ML ensemble (Isolation Forest + Gradient Boosted Trees) with graph-based fraud-ring clustering; diagnosed and fixed a three-stage bug cascade in ring detection, reaching 100% precision.
- Found a critical data-leakage bug via live API testing that the offline benchmark had missed — the model had learned "new account = fraud" as a shortcut (85.5% of decision weight); retrained to cut it to 4.4% and correctly catch a real account-takeover case that had previously slipped through.
- Designed a schema-bound LLM reasoning layer (Pydantic-validated output, deterministic fail-safe fallback) that engages only for ambiguous risk scores, plus a custom "grounding check" that catches fabricated LLM evidence citations — a failure mode schema validation alone cannot catch.
- Found and fixed a concurrency race condition that silently forked the tamper-evident audit ledger under simultaneous load, and closed a prompt-injection delimiter-spoofing vulnerability using randomized per-request boundary tokens.
- Verified a real, working MCP (Model Context Protocol) integration end-to-end — the same architecture Razorpay's own Agent Studio is built on — and applied Clopper-Pearson statistical bounds to honestly report real-world precision at true fraud base rates.
- Built a 21-test adversarial regression suite; caught a submission-breaking hardcoded-path bug via clean-room testing and cross-verified every fix on independent Windows hardware.

300 Days of AI/ML — *Day 191/300, in progress* · [Repo](#)

- Public, daily-tracked ML learning log spanning data foundations through deep learning, including coursework-based assignments and personal projects.
- Completed the full unsupervised learning phase with a SmartCart customer segmentation pipeline (K-Means/DBSCAN/hierarchical clustering) on 2,240 customers, identifying 4 actionable segments plus 94 VIP outliers.
- Now in the Deep Learning (PyTorch) phase: built two ANNs from scratch — 93.88% accuracy on a 7-class classification task, and $R^2 = 0.937$ on a regression task.

ClassroomAI Monitor — *1st Place, BYTEHACK 2026* · [Repo](#)

- Built a real-time classroom engagement system combining YOLOv8 (person detection), MediaPipe (head-pose and gesture tracking), and OpenAI Whisper (live transcription) into one live pipeline.
- Sole author of the full stack — FastAPI backend, React dashboard, and the AI pipeline itself — under a hackathon deadline, with an on-device privacy mode (face blur, no raw video stored).

CampusGigs — *1st Place, Software Project, National Youth Festival 2026* · [Repo](#)
[Live App](#)

- Built the full React frontend and the prompt-engineering logic behind an AI gig-description generator (Google Gemini) for a peer-to-peer student gig marketplace.
- Also designed a simulated UPI wallet and real-time negotiation chat. Team DragonDevZ — Lead Web Developer & Prompt Engineer.

AI ATS Resume System — *active development* · [Repo](#)

- Built a hybrid resume-to-job-description matcher combining TF-IDF/BM25 lexical matching, semantic embeddings, a trained shortlist-prediction classifier, and an LLM feedback layer.
- FastAPI backend, React frontend, Dockerized, with Supabase auth and PDF report export.

Kaggriculture Agent — *active, Kaggle competition (\$50K prize pool)* · [Repo](#)

- Building an autonomous farming agent competing head-to-head over a 720-turn season, managing crops, labor, land, and a dynamic market economy.
- Validated against the real kaggle-environments engine, with an evolving strategy layer covering labor scaling, multi-unit task coordination, and market-throttling logic.

Memento Mori Life Tracker — *active development* · [Repo](#)

- Building a premium, offline-first life-tracking PWA (React, Tailwind CSS, Vite) featuring a 60-year life grid and full PWA offline capabilities.

COMPETITIONS & ACHIEVEMENTS

- **Open Source Connect India 2026** — Selected as an Official Contributor (Sept 2026)
- **BYTEHACK 2026, BCE Patna** — 1st Place — ClassroomAI Monitor
- **RookieCTF 2026 (HackerShield, Tatva Club, IIT Patna)** — 1st Place — Crypto / Forensics / OSINT / Web, 3,648 pts
- **29th National Youth Festival — Science Mela 2026, BCE Patna (Govt. of Bihar)** — 1st Place, Software Project — CampusGigs
- **Adobe University Hackathon 2026** — Round 1 cleared — 26,760 selected out of 3,01,082 registered teams (top ~9%)
- **Razorpay AI Buildathon 2026** — Submitted Sept 5, 2026 (result pending) — Sentinel
- **Kaggle — Kaggriculture** — Active competitor, \$50,000 prize pool

EDUCATION

BS-MS, Artificial Intelligence & Cyber Security — IIT Patna | CGPA: 7.53 | 2nd Semester, Jan 2026 – 2031

B.Tech, Computer Science & Engineering (IoT) — BCE Patna | CGPA: 8.15 | 3rd Semester, Aug 2025 – 2029